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Lens cap assembly

Abstract

Disclosed are implementations of a lens cap assembly, each configured to shroud the lens of a device such as a flashlight, an optical sight, or a night vision device. Each lens cap assembly comprises a sleeve, either a removable lens or an anti-reflection element positioned within the sleeve, and a cap. The cap is removably secured to the sleeve by an elastic cord.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) (1) This application claims the benefit of U.S. Provisional Application Ser. No. 63/523,015, filed on Jun. 23, 2023, the entirety of which is incorporated herein by reference.

TECHNICAL FIELD

(1) This disclosure relates to implementations of a lens cap assembly.

BACKGROUND

(2) The lens of a flashlight, telescopic sight, or night vision device can be damaged by dust, debris, and impacts, which can scratch or crack the lens. Additionally, the highly reflective head of a weapon light, as well as the lens of a telescopic sight or night vision device, can reflect light that reveals the user's location. This is particularly concerning for users trying to conceal their position, such as warfighters.

(3) In the case of a flashlight, protecting against accidental light discharges is crucial, especially for warfighters operating under night vision, where an unintended white light discharge could compromise a mission or endanger personnel. Therefore, it is important to safeguard these devices from physical damage and prevent reflections and accidental light emissions that could reveal the user's presence.

(4) Accordingly, there exists a need for the lens cap assembly disclosed herein. The present invention is primarily directed towards providing a lens cap assembly configured to address these and other needs.

SUMMARY OF THE INVENTION

(5) It is to be understood that this summary is not an extensive overview of the disclosure. This summary is exemplary and not restrictive, and it is intended neither to identify key or critical elements of the disclosure nor delineate the scope thereof. The sole purpose of this summary is to explain and exemplify certain concepts of the disclosure as an introduction to the following complete and extensive detailed description.

(6) Disclosed are implementations of a lens cap assembly, each configured to shroud the lens of a device such as a flashlight, an optical sight, or a night vision device. Each lens cap assembly comprises a sleeve, either a removable lens or an anti-reflection element positioned within the sleeve, and a cap. The cap is removably secured to the sleeve by an elastic cord.

(7) An example lens cap assembly comprises a sleeve configured to frictionally engage with a flashlight head, a removable lens, and a cap. The sleeve has a bore disposed through it. A front end of the bore includes a radially inwardly projecting ledge defining a circular opening having a diameter that is less than a diameter of the bore. A front end of the sleeve includes an annular chamfer positioned around the circular opening defined by the radially inwardly projecting ledge. The removable lens is positioned within the bore against the radially inwardly projecting ledge of the sleeve. The cap is removably secured to the front end of the sleeve by an elastic cord. The cap includes a backside and an annular chamfer disposed on the backside configured to interface with the annular chamfer on the front end of the sleeve.

(8) Another example lens cap assembly comprises a sleeve configured to frictionally engage with an optical sight, an anti-reflection element configured to filter out light reflections from a lens of the optical sight, and a cap. The sleeve has a bore disposed through it. A front end of the sleeve includes an annular chamfer positioned around an exit opening of the bore. The anti-reflection element is positioned within the bore of the sleeve. The cap is removably secured to the front end of the sleeve by an elastic cord. The cap includes a backside and an annular chamfer disposed on the backside configured to interface with the annular chamfer on the front end of the sleeve.

(9) Yet another example lens cap assembly comprises a sleeve configured to frictionally engage with an objective end of a night vision device, a removable lens, a cap, and a sliding member. The sleeve has a bore disposed through it. The front end of the bore includes a radially inwardly projecting ledge defining a circular opening having a diameter that is less than a diameter of the bore. The front end of the sleeve includes an annular chamfer positioned around the circular opening defined by the radially inwardly projecting ledge. The removable lens is positioned within the bore against the radially inwardly projecting ledge of the sleeve. The cap is removably secured to the front end of the sleeve by an elastic cord. The cap includes a backside, an annular chamfer disposed on the backside configured to interface with the annular chamfer on the front end of the sleeve, and an aperture positioned to allow light to an objective lens of the night vision device. The

sliding member includes at least one aperture. The sliding member is configured to move between a first position in which the sliding member covers the aperture of the cap and a second positioned in which the at least one aperture of the sliding member overlays the aperture of the cap. The aperture of the cap and the at least one aperture of the sliding member are configured together to change a focal distance of the night vision device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 illustrates an isometric view of a flashlight equipped with a lens cap assembly according to the principles of the present disclosure.
- (2) FIG. 2 illustrates an isometric view of the lens cap assembly shown in FIG. 1.
- (3) FIG. 3 illustrates an exploded view of the lens cap assembly shown in FIG. 2.
- (4) FIG. 4 illustrates a right side elevational view of the lens cap assembly shown in FIG. 2.
- (5) FIG. 5 illustrates a top plan view of the lens cap assembly shown in FIG. 2.
- (6) FIG. 6 illustrates a cutaway view of the lens cap assembly taken along line A-A shown in FIG. 5.
- (7) FIG. 7 illustrates another top plan view of the lens cap assembly shown in FIG. 2.
- (8) FIG. 8 illustrates a front elevational view of the lens cap assembly shown in FIG. 2.
- (9) FIG. 9 illustrates a cutaway view of the lens cap assembly taken along line B-B shown in FIG. 8.
- (10) FIG. 10 illustrates a telescopic sight equipped with another lens cap assembly according to the principles of the present disclosure.
- (11) FIG. 11 illustrates an isometric view of the lens cap assembly shown in FIG. 10.
- (12) FIG. 12 illustrates an exploded view of the lens cap assembly shown in FIG. 11.
- (13) FIG. 13 illustrates a right side elevational view of the lens cap assembly shown in FIG. 11.
- (14) FIG. 14 illustrates a front elevational view of the lens cap assembly shown in FIG. 11.
- (15) FIG. 15 illustrates a cutaway view of the lens cap assembly taken along line A-A shown in FIG. 14.
- (16) FIG. 16 illustrates a top plan view of the lens cap assembly shown in FIG. 11.
- (17) FIG. 17 illustrates another front elevational view of the lens cap assembly shown in FIG. 11.
- (18) FIG. 18 illustrates a cutaway view of the lens cap assembly taken along line B-B shown in FIG. 17.
- (19) FIG. 19 illustrates a night vision device equipped with yet another lens cap assembly according to the principles of the present disclosure.
- (20) FIG. 20 illustrates an isometric view of the lens cap assembly shown in FIG. 19.
- (21) FIG. 21 illustrates an exploded view of the lens cap assembly shown in FIG. 20.
- (22) FIG. 22 illustrates a right side elevational view of the lens cap assembly shown in FIG. 20.
- (23) FIG. 23 illustrates a front elevational view of the lens cap assembly shown in FIG. 20.
- (24) FIG. 24 illustrates a cutaway view of the lens cap assembly taken along line A-A shown in FIG. 23.
- (25) FIG. 25 illustrates a top plan view of the lens cap assembly shown in FIG. 20.
- (26) FIG. 26 illustrates another front elevational view of the lens cap assembly shown in FIG. 20.
- (27) FIG. 27 illustrates a cutaway view of the lens cap assembly taken along line B-B shown in FIG. 26.
- (28) Like reference numerals refer to corresponding parts throughout the several views of the drawings.

DETAILED DESCRIPTION

- (29) FIGS. 1-9 illustrate an example lens cap assembly 100 configured for attachment to a

flashlight **102**. As shown in FIG. 1, the lens cap assembly **100** is configured to shroud the head **104** of the flashlight **102** and thereby protect the lens and lamp assembly of the flashlight **102**.

(30) As shown best in FIGS. 2 and 3, the lens cap assembly **100** comprises a sleeve **110**, a removable lens **112**, and a cap **114**. The removable lens **112** is press fit into position within the sleeve **110** and the cap **114** is attached to the sleeve **110** by an elastic cord **116**.

(31) The sleeve **110** is configured to frictionally engage with the flashlight head **104**. A bore **118** is disposed through the sleeve **110**; the diameter of the bore **118** will be only slightly larger than the diameter of the flashlight head **104** to which the lens cap assembly **100** is being attached, to allow for a tight, frictional engagement while retaining removability. A front end of the bore **118** includes a radially inwardly projecting ledge **124** that defines a circular opening **126**, which has a diameter smaller than the diameter of the bore **118** (see, e.g., FIG. 6). A front end of the sleeve **110** includes an annular chamfer **128** positioned around the circular opening **126** defined by the radially inwardly projecting ledge **124** (see, e.g., FIG. 3). The sleeve **110** also includes two external tabs **120** that extend laterally from opposite sides. Each tab **120** includes a thru hole **122**, the purpose of which will be described in detail below. The sleeve **110** is made of polyamide 12 (PA12). However, the sleeve **110** can be made of other suitable materials, such as a thermoplastic polyurethane.

(32) As shown in FIGS. 6 and 9, the removable lens **112** is positioned within the bore **118** against the radially inwardly projecting ledge **124** of the sleeve **110**. Preferably, the lens **112** is removably and frictionally engaged within the bore **118** of the sleeve **110**. Therefore, the diameter of the lens **112** is only slightly smaller than the diameter of the bore **118** in which the lens **112** is being positioned, to allow for a tight, frictional engagement while retaining removability. The removable lens **112** is disc-shaped and preferably made of a transparent, substantially rigid material such as glass. However, the lens **112** can be made of other suitable materials, such as polycarbonate or acrylate polymer. In some implementations, the removable lens **112** may be configured to limit the color of a light beam emitted by the flashlight **102** to a particular color, such as blue or red.

(33) As shown in FIG. 1, the cap **114** is removably secured to the front end of the sleeve **110** by the elastic cord **116**. An annular chamfer **134** disposed on a backside of the removable cap **114** is configured to interface with the annular chamfer **128** on the front end of the sleeve **110** (see, e.g., FIG. 6). Together, the two chamfers (**128** and **134**) serve to center the cap **114** on the front end of the sleeve **110**. In this way, the cap **114** is positioned to cover the removable lens **112** held by the sleeve **110**. The cap **114** includes two external tabs **130** that extend laterally from opposite sides. Each tab **130** includes a thru hole **132**, the purpose of which will be described in detail below. The cap **114** also includes a flipper tab **136** that extends laterally from the cap **114**. The flipper tab **136** is configured so that it can be used to dislodge the cap **114** from the front end of the sleeve **110**. The cap **114** is made of a thermoplastic polyurethane (TPU). However, the cap **114** can be made of other suitable materials, such as polyamide 12 (PA12).

(34) Though the example cap **114** includes one flipper tab **136**, in other implementations, the cap **114** may include two flipper tabs, each of which extend laterally from the cap **114**.

(35) As previously disclosed, the elastic cord **116** is used to removably secure the cap **114** to the front end of the sleeve **110**. The elastic cord **116** is a length of 2.5 mm shock cord, though another thickness could be used. Each end **138**, **140** of the elastic cord **116** extends through a pair of tabs, one external tab **120** on the sleeve **110** and one external tab **130** on the cap **114**. Each end **138**, **140** of the elastic cord **116** includes a stopper knot **142**, preferably an overhand knot, that prevents the end of the elastic cord **116** from unreeving. When the cap **114** is dislodged from the front end of the sleeve **100**, the elastic cord **116** is configured to draw the cap **114** against the side of the lens cap assembly **100**.

(36) FIGS. 10-18 illustrate another example lens cap assembly **200**. This lens cap assembly **200** is configured for attachment to the objective end of a telescopic sight **202**. The lens cap assembly **200** is similar to the lens cap assembly **100** discussed above, but the sleeve **210** includes an anti-reflection element **212** instead of a removable lens **112** and a radially inwardly projecting ledge

124.

(37) The anti-reflection element **212** is configured to filter out light reflections from the objective lens of the telescopic sight **202**. The anti-reflection element **212** comprises a plurality of honeycomb shaped apertures **214** positioned in-line with the bore **218** within the sleeve **210** (see, e.g., FIG. 12). However, the apertures **214** of the kill flash **212** could be square, octagonal, diamond, and the like. The plurality of apertures **214** that make up the anti-reflection element **212** are an integral portion of the sleeve **210**.

(38) The diameter of the bore **218** in the sleeve **210** will be only slightly larger than the diameter of the objective end of the telescopic sight **202** to which the lens cap assembly **200** is being attached, to allow for a tight, frictional engagement while retaining removability.

(39) FIGS. 19-27 illustrate yet another example lens cap assembly **300**. This lens cap assembly **300** is configured for attachment to the objective end of a night vision device **302**. The lens cap assembly **300** is similar to the lens cap assemblies **100**, **200** discussed above, in particular the lens cap assembly **100** shown in FIGS. 1-9, but the cap **314** includes an aperture **316** and a sliding member **318** that together can be used to change the focal distance of the night vision device **302**. Specifically, the user may optionally toggle the sliding member **318** between a closed state and one of two open states. Each of the open states changes the focal distance of the night vision device **302**.

(40) The aperture **316** in the cap **314** is positioned to correspond with the approximate center of the objective lens of the night vision device **302**. The aperture **316** is circular and has a 6 mm diameter. While the example aperture **316** has a 6 mm diameter, the diameter of the aperture **316** can vary so long as the aperture **316**, at a minimum, has the same diameter as the largest aperture in the sliding member **318** discussed in detail below.

(41) The sliding member **318** is configured (e.g., shaped and dimensioned) to slide within a guide slot **344** formed between a pair of arms **346** on a front side of the cap **314**. The two arms **346** are positioned on opposite sides of the aperture **316**, each having a lip **348** extending partially inward from opposing sides of the aperture **316** (see, e.g., FIG. 20).

(42) The sliding member **318** comprises an elongate body **320** having a stop (**322** and **324**) on each end configured to limit the movement of the sliding member **318** between first, second and third positions. One of the stops **324** is a removable lug secured to an end of the elongate body **320** by a threaded fastener **326** and a nut **328** (see, e.g., FIG. 24). The bottom of the lug **324** includes a cutout configured to receive a mating protuberance **330** adjacent an end of the elongate body **320**. The shape of the protuberance **330** prevents the lug **324** from rotating.

(43) The elongate body **320** includes two apertures, a first aperture **332** configured to provide for a focal distance of approximately 1-5 feet and a second aperture **334** configured to provide for a focal distance of approximately 10 feet. In the preferred implementation, the first aperture **332** is 3 mm in diameter and the second aperture **334** is 5 mm in diameter. However, depending on the desired focal distance for the night vision device **302**, an aperture may have a diameter greater than or less than what is listed above.

(44) A bump **336**, or rounded protuberance, is disposed on the backside of the sliding member **318** (see, e.g., FIG. 24). The bump **336** is configured to frictionally engage with one of three detents **338** in the front side of the cap **314**. The rounded shape of the bump **336** allows for smooth movement and engagement, while still providing enough friction to prevent unintended sliding of the sliding member **318**. The detents **336** are positioned in-line with the aperture **316** in the cap **314** (see, e.g., FIG. 21).

(45) The sliding member **318** slides within the guide slot **344** between the first, second, and third positions. In the first position, the sliding member **318** covers the aperture **316** in the cap **314**. In the second position, the first aperture **332** in the sliding member **318** overlays the aperture **316** in the cap **314** (see, e.g., FIG. 24). In the third position, the second aperture **334** in the sliding member **318** overlays the aperture **316** in the cap **314**. Moving the sliding member **318** between the second

and third positions changes the focal distance of the night vision device **302**.

(46) Though the example sliding member **318** includes two apertures (**332** and **334**), in other implementations, the sliding member **318** may have no aperture, a single aperture, or more than two apertures. The limit to the number of apertures being a function of aperture diameter and/or the length of the elongate body **320**.

(47) The foregoing description of the invention is intended to be illustrative; it is not intended to be exhaustive or to limit the claims to the precise forms disclosed. Those skilled in the relevant art can appreciate that many modifications and variations are possible in light of the foregoing description and associated drawings.

(48) Reference throughout this specification to an “embodiment” or “implementation” or words of similar import means that a particular described feature, structure, or characteristic is included in at least one embodiment of the present invention. Thus, the phrase “in some implementations” or a phrase of similar import in various places throughout this specification does not necessarily refer to the same embodiment.

(49) The described features, structures, or characteristics may be combined in any suitable manner in one or more embodiments. In the above description, numerous specific details are provided for a thorough understanding of embodiments of the invention. One skilled in the relevant art will recognize, however, that embodiments of the invention can be practiced without one or more of the specific details, or with other methods, components, materials, etc. In other instances, well-known structures, materials, or operations may not be shown or described in detail.

Claims

1. A lens cap assembly for an optical sight, the lens cap assembly comprising: a sleeve configured to frictionally engage with the optical sight, the sleeve having has a bore disposed through it, wherein a front end of the sleeve includes an annular chamfer positioned around an exit opening of the bore; an anti-reflection element configured to filter out light reflections from a lens of the optical sight, the anti-reflection element being positioned within the bore of the sleeve; and a cap removably secured to the front end of the sleeve by an elastic cord, wherein the cap includes a backside and an annular chamfer disposed on the backside configured to interface with the annular chamfer on the front end of the sleeve.
 2. The lens cap assembly of claim 1, wherein the sleeve includes two external tabs that extend laterally from opposite sides of the sleeve, each of the two external tabs includes a thru hole; the cap includes two external tabs that extend laterally from opposite sides of the cap, each of the two external tabs includes a thru hole; wherein the elastic cord extends through the holes of the external tabs of the sleeve and the cap.
 3. The lens cap assembly of claim 2, wherein the cap includes a flipper tab that extends laterally from the cap, the flipper tab is configured to dislodge the cap from the front end of the sleeve.
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